

DATA ENGINEER INTERN

TRAN THIEN PHUC

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SUMMARY

Data Engineering student with hands-on experience building batch and real-time data pipelines, processing time-series and multimedia data, and designing reproducible data workflows. Experienced in Python, SQL, Kafka, PySpark, Docker, QuestDB, data ingestion, transformation, validation, and AI-assisted development workflows. Familiar with prompt engineering and applying generative AI tools to accelerate development, data exploration, documentation, and problem-solving. Seeking a Data Engineering internship to apply my data processing and engineering skills to reliable, scalable, and production-oriented data systems.

EDUCATION

Ho Chi Minh City University of Technology and Engineering (HCM-UTE)

Bachelor of Data Engineering

Sep 2023 – Present

Ho Chi Minh City, Vietnam

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript/TypeScript.

Data Engineering: Kafka, Data Ingestion, ETL Pipelines, Batch Processing, Stream Processing, Data Validation, Backfill/Reconciliation, Time-series Data.

Data Processing: Pandas, NumPy, PyArrow, PySpark, OpenCV, MediaPipe.

Databases: PostgreSQL, MySQL, QuestDB. Supabase

Backend & APIs: NestJS, REST APIs, Socket.IO, Swagger.

BI & Visualization: Power BI.

AI & Machine Learning: Scikit-learn, PyTorch, Transformer Models, Prompt Engineering, Generative AI Applications, AI-assisted Development.

Tools: Git, Docker, Docker Compose, WSL, Jupyter Notebook, Google Colab, Excel.

PROJECTS

NexTick - Real-time Cryptocurrency Market Data Pipeline | *Python, Kafka, QuestDB, Docker*

2026

- Built an end-to-end real-time data pipeline that ingests raw **Binance trade events**, publishes them to **Kafka**, aggregates OHLCV candles in Python, and persists finalized time-series data in **QuestDB**.
- Implemented a startup **backfill and reconciliation workflow** to recover missing historical candles and establish a UTC cutover between historical and live processing, preventing overlapping data writes.
- Designed event data contracts for raw trades and candle streams and supported multiple trading symbols and time intervals across ingestion, processing, storage, and downstream consumers.
- Containerized Kafka, Kafka UI, QuestDB, producer, backfill, and stream-processing services with **Docker Compose** for reproducible local deployment and orchestration.
- Developed a **NestJS** data-serving layer that queries historical candles from QuestDB, consumes real-time Kafka events, and exposes data through REST APIs and Socket.IO.
- Source:** github.com/iamzaneth/NexTick

WLASL Sign Language Data Pipeline | *Python, MediaPipe, NumPy, PyTorch*

2026

- Built an end-to-end data preparation pipeline for the **WLASL** dataset, covering raw video auditing, official split construction, landmark extraction, preprocessing, feature generation, and experiment-ready artifacts.
- Implemented dataset validation and video-readability auditing while preserving official **WLASL100/300/1000/2000** splits and filtering them to locally available, usable samples.
- Developed resumable **MediaPipe** landmark extraction for pose, hands, and facial regions, with schema validation, shoulder-centered normalization, and short-gap hand landmark recovery.
- Built a reusable **memory-mapped fixed-length feature cache** to reduce repeated preprocessing and efficiently provide prepared sequence data to downstream model-training workloads.

- Generated reproducible JSON/CSV experiment metadata and integrated the processed dataset with a Transformer encoder training pipeline; the tracked WLASL100 cache contains **1,013 usable videos**.
- **Source:** github.com/iamzaneth/wlasl-sign-language-recognition

RELEVANT COURSEWORK

Data Structures and Algorithms, Database Systems, Machine Learning, Data Mining, Big Data Processing, Statistics, Web Programming.

ADDITIONAL INFORMATION

Languages: Vietnamese, English.

Career Interests: Data Engineering, Data Platforms, Streaming Systems, Data Infrastructure, AI-powered Data Applications.